

Andrzej Fijało

Programmer, R&D / MLOps Engineer

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Education

MSc Gdańsk University of Technology, Computer Science Gdańsk
Eng. Faculty of Electronics, Telecommunications and Informatics 2023 – 2025
Specialization: **Mobile Systems and Technologies**

Eng. Gdańsk University of Technology, Automation and Robotics Gdańsk
Faculty of Electrical and Control Engineering 2018 – 2022
Profile: **Automation and Robotics**

Work Experience

R&D Engineer, Software Engineer / MLOps Gdynia
Advanced Protection Systems 2022 – present

Work on autonomous systems, robotics, and image processing for unmanned systems, including development of MLOps and ML pipelines.

- Work on an interceptor drone for neutralizing airborne objects.
- Development and optimization of image processing pipelines (object detection, visual odometry).
- Designing and optimizing video pipelines in GStreamer and DeepStream on NVIDIA Jetson platforms.
- Software-side integration of sensors, including thermal cameras, and data collection for algorithm development and validation.
- Work with robotic systems and ROS in embedded environments.
- Designing processing pipelines for sensor and vision data.
- Took ownership of video detection development within the company.
- Integration of AI solutions (YOLO, TensorRT, OpenCV) in embedded systems.
- Development of custom video tracking applications using DeepStream SDK.
- Data and model versioning, CVAT usage for dataset annotation and management.
- Refactoring codebase to Kedro + MLflow, configuring external logging, edge-AI deployment, model versioning and result visualizations.
- Building self-hosted MLOps using Docker and MLflow for tracking model training.
- Hardware debugging and integration of new sensors.
- Analysis and optimization of system resource usage (CPU, GPU, VRAM).
- Creating developer tools to improve team workflow.
- Technology research and preparing technical documentation.
- Mentoring interns.

Product Development Team Lead Gdańsk
VAVE Light 2018 – 2022

Development of technology products in a small engineering team.

- Designing and bringing new products from concept to production.
- Optimizing manufacturing processes (assembly, testing, quality control).
- Designing enclosures and selecting electronic and mechanical components.
- Performing functional and environmental product testing.
- Creating technical and production documentation.
- Collaborating with suppliers and subcontractors on prototyping and production.
- Analyzing customer requirements and adapting products.

Skills

Programming & Tools: C++, Python, Bash, Git, Nix, Docker, CMake, uv, Jenkins

Image Processing: OpenCV, GStreamer, DeepStream, YOLO, CUDA, VPI, FFmpeg, thermal cameras

Robotics & Embedded Systems: NVIDIA Jetson, Linux, ROS, V4L2, thermal cameras, OpenFOAM

Machine Learning & MLOps: PyTorch, MLflow, Kedro, TensorRT

DevOps & Infrastructure: Docker, NixOS, MLflow self-hosted, Nginx, Auth0/OIDC, edge-AI deployment

Natural Languages: polish (native), english (B2)